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Can Decision Models Understand Stance ?

Evaluating Jev Against General Purpose LLMs

Authors: Xing Li, Jinzhong Ning, Yijia Zhang, Liang Yang, Hongfei Lin
Dalian Maritime University
Dalian University of Technology

Presented By :
Hema Meena R

What is Stance Detection ?

Stance detection is the task of identifying a person's position or opinion toward a particular target.

The paper considers three possible stance labels:

- Favor: The text supports or agrees with the target.
- Against: The text opposes or disagrees with the target.
- Neutral: The text does not clearly support or oppose the target.

Example:

Target: Online education is beneficial.

- “It provides flexible learning.” → Favor
- “It reduces direct interaction.” → Against
- “Online education is widely used.” → Neutral

FAVOR

Supports or agrees with the target

AGAINST

Opposes or criticizes the target

NEUTRAL

No clear position towards the target

Why is Stance Detection Challenging?

-Identifying a person's stance is not always straightforward.

Context matters

The meaning of a statement may depend on earlier messages.

Indirect opinions

People may express agreement or disagreement without using direct words.

Different targets

A statement may support one idea but oppose another.

Ambiguous statements

Some text may not clearly express any position.

CONVERSATION UNDERSTANDING

Speaker A : “Online education is very useful”

Speaker B : “I completely disagrees”

The speaker and conversation history may be necessary to interpret the actual opinion.

The earlier message helps to interpret the reply

Research Motivation and Objective

Why did the researchers conduct this study?

- Traditional stance detection approaches often depend on task-specific labelled datasets.
- General-purpose Large Language Models (LLMs) can also perform stance classification using instructions.
- Stance detection has three predefined output labels: Favor, Against and Neutral.
- The researchers investigate whether Jev can perform this classification effectively compared with general-purpose LLMs.

Related Work : Stance Detection

1. **Target-specific stance detection**
   - Identifies the relationship between a text and a particular target.
   - Example: Understanding whether a text supports or opposes a given topic.
2. **Cross-target stance detection**
   - Transfers knowledge between related targets.
   - Reduces the need for labelled data for every target.
3. **Zero-shot stance detection**
   - Identifies stance toward targets that were not seen during training.
   - Example: VAST dataset.
4. **LLM-based approaches**
   - Use Large Language Models to identify relationships between text and target.
   - Some approaches use LLM-generated knowledge to support stance prediction.

Related Work : Conversational Stance Detection

What is Conversational Stance Detection (CSD)?

Conversational stance detection identifies a speaker's position toward a target by considering the conversation and interactions between speakers.

Important research works:

- SRQ: Studies stance in replies and quoted tweets.
- MT-CSD: Models dependencies across multiple conversation turns.
- GLAN: Combines global and local conversational information.
- ZS-CSD: Introduces a Chinese benchmark for stance detection toward unseen targets.
- SITPCL: Combines speaker interaction with target-aware contrastive learning.

WorkFlow

DATASETS

VAST
(Individual text)

ZS-CSD
(Conversational)

INPUT PREPARATION

- Target
- Focus Text
- Conversation History
- Target Type

Jev-Choice Based Decision

Task Instruction

Determine the stance towards the target

Three Choices

FAVOR
Supports or agrees with the target

AGAINST
Opposes or criticizes the target

NEUTRAL
No clear position towards the target

Jev
Select One Option

Prediction Extraction

Extract & Validate Selected Label

- FAVOR
- AGAINST
- NEUTRAL

Evaluation Metrics

Metrics

- F1 Score
- Accuracy

Different Stages

1

TASK DEFINITION



Identify the stance of the focus text towards a given target.

FAVOR

AGAINST

NEUTRAL

2

STRUCTURED INPUT CONSTRUCTION



- Prepare the input using:
  - Target
  - Focus text
  - Conversation history
  - Target type
- Preserve the original language and conversation order.

3

JEV CHOICE BASED DECISION

FAVOR

AGAINST

NEUTRAL

- Provide the task instruction and three stance options to Jev.
- Jev selects exactly one option based on the given criteria:
  - Favor – Support or approval
  - Against – Opposition or criticism
  - Neutral – No clear stance

4

PREDICTION EXTRACTION



- Extract the selected option from Jev's response.
- Validate the selected label.
- Use it as the final stance prediction.

Complete Example : Input To Prediction

1. **Target**

   Online education is better than classroom education

2. **Conversation History**

   Speaker A : “Online education gives students more flexibility.”

3. **Focus Text**

   Speaker B : “I completely disagree. Students learn better through direct classroom interaction.”



Predicted

AGAINST

Datasets , Models and Evalution Metrics

1. **Datasets**
   - VAST: Individual English text – 3,006 test records.
   - ZS-CSD: Chinese conversational text – 2,584 test records; focuses on unseen-target stance detection.
2. **Models**
   - Main Model: Jev
   - General-Purpose LLMs:
     DeepSeek V4.1 Flash, Qwen3.8-Flash, GPT-5.6, Claude Sonnet 5.
   - Fine-Tuned Reference Models: Qwen2.5-3B (LoRA), Qwen2.5-7B (LoRA).
3. **Evaluation Metrics**
   - **Accuracy:** Measures the overall percentage of correct predictions.
   - **Macro-F1 Score:** Measures performance across Favor, Against, and Neutral classes.
   - **Token Usage:** Measures the number of tokens consumed.
   - **API Cost:** Measures the cost of processing the dataset.

Results

Dataset	Model / setting	Accuracy	Macro-F1	Input tokens	Output tokens	Cost
VAST	Jev	78.14	77.92	1,763,143	114,228	$0.0741
	DeepSeek V4.1 Flash	74.82	73.99	865,478	18,708	$0.2133
	Qwen3.8-Flash	74.52	73.79	763,609	28,874	$0.1021
	GPT-5.6	**78.51**	**78.08**	1,178,659	33,439	$3.9010
	Claude Sonnet 5	74.82	74.51	2,279,907	251,109	$4.7051
	Qwen2.5-3B-Instruct (SFT)†	74.16	74.52	—	—	—
	Qwen2.5-7B-Instruct (SFT)†	79.32	79.91	—	—	—
ZS-CSD	Jev	58.40	58.43	1,846,931	98,192	$0.0776
	DeepSeek V4.1 Flash	58.28	56.89	815,813	15,920	$0.1829
	Qwen3.8-Flash	54.61	53.16	707,481	27,236	$0.0948
	GPT-5.6	**74.65**	**73.62**	1,140,780	35,592	$3.9230
	Claude Sonnet 5	72.52	72.42	2,402,197	325,790	$5.5582
	Qwen2.5-3B-Instruct (SFT)†	55.76	56.08	—	—	—
	Qwen2.5-7B-Instruct (SFT)†	60.98	60.85	—	—	—

Table 1: Performance and resource usage on the full test sets of VAST and ZS-CSD. Bold indicates the best results among prompted models. Dashes denote unavailable measurements. †SFT results are from single runs without per-instance predictions.

Results

Dataset	Model	Against	Favor	Neutral
VAST	Jev	77.13	72.71	**83.92**
	DeepSeek	74.15	66.83	81.00
	Qwen	73.88	65.52	81.97
	GPT-5.6	**78.25**	**73.95**	82.05
	Claude	75.35	70.26	77.91
ZS-CSD	Jev	56.66	59.58	59.04
	DeepSeek	54.60	51.18	**64.89**
	Qwen	55.59	42.45	61.42
	GPT-5.6	**79.49**	**82.70**	58.67
	Claude	75.78	77.50	63.99

Table 2: Per-class F1 (%). Bold indicates the best result for each class within each dataset.

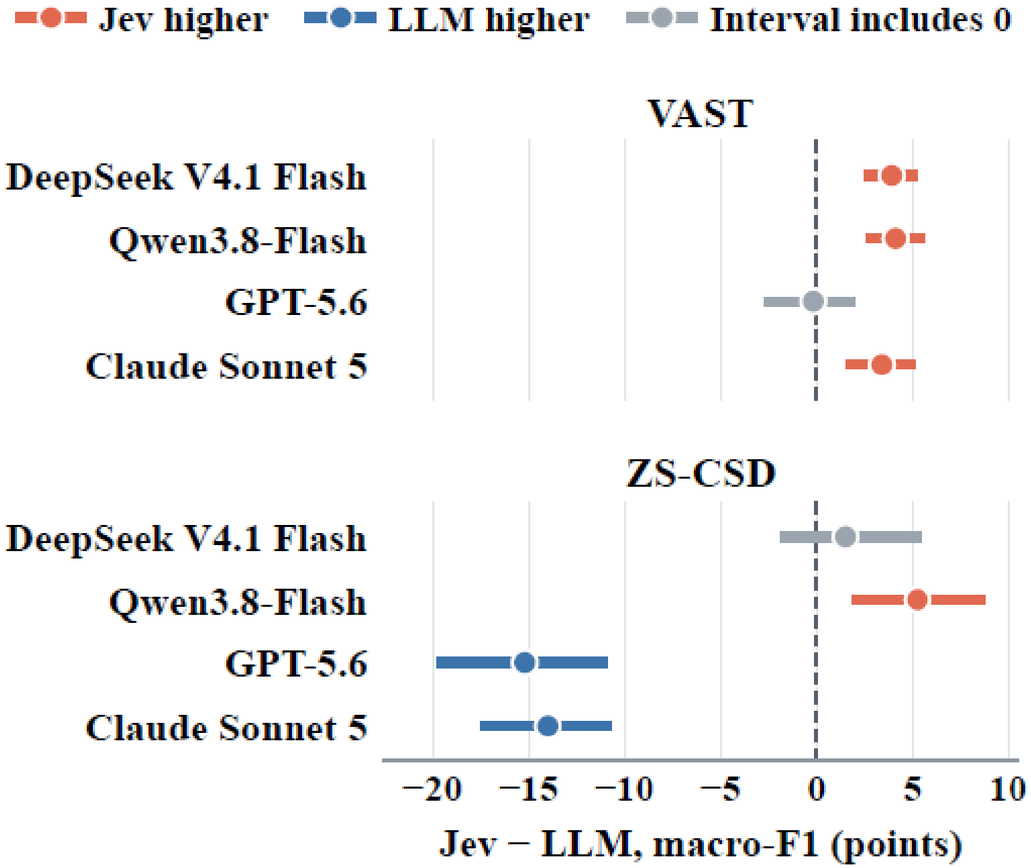


Figure 1: Macro-F1 differences between Jev and each LLM (Jev minus LLM, in percentage points), with paired target-bootstrap 95% intervals.

Why Does Jev Perform Differently?

1. **Individual Text vs. Conversations**
   - VAST: 77.92% Macro-F1 – competitive performance on individual English text.
   - ZS-CSD: 58.43% Macro-F1 – lower performance on Chinese conversational text.
2. **Difficulty in Identifying Stance Direction**
   - Jev makes 373 Favor–Against classification errors.
   - GPT-5.6 makes 35; Claude Sonnet 5 makes 101.
   - Indicates difficulty interpreting support and opposition between speakers.
3. **Target Complexity**
   - Noun Phrase: 63.49% Macro-F1.
   - Claim: 55.26% Macro-F1.
   - Claims may contain complex semantics, negation and qualifying conditions.
4. **Conversation Length**
   - Performance gap does not consistently increase with conversation length.
   - Reply relationships and stance direction may be more important than the number of turns.

Real-World Applications & Suitability

1. **Social Media Opinion Monitoring**
   - Classify public opinions towards products, topics or policies.
   - Track Favor, Against and Neutral opinions in incoming posts.
2. **Customer Feedback Analysis**
   - Identify whether customers support or oppose a product feature, service change or company decision.
   - Help businesses monitor customer reactions.
3. **Public Opinion & Policy Analysis**
   - Analyse citizens' positions towards proposed policies, public initiatives or social issues.
   - Monitor changes in public support and opposition.
4. **Online Campaign Monitoring**

-Track audience support or opposition towards campaign messages, proposals and announcements.

Real-World Applications & Suitability

More Promising

1. **Individual-text opinion classification**

77.92 VAST-Macro-F1

Classifying Favor, Against and Neutral opinions in individual English text, such as social media posts.

Relatively Better

2. **Stance toward named entities**

63.49 ZS-CSD Noun Phrase Macro-F1

Identifying whether a speaker supports, opposes or is neutral toward a person, organization or product mentioned in a conversation.

Limited suitablility

3. **Conversational stance classification**

58.43 ZS-CSD Overall Macro-F1

Classifying participants' Favor, Against and Neutral positions in Chinese conversational discussions

More Challenging

4. **Complex claim-based stance detection**

55.26 ZS-CSD Claim Macro-F1

Identifying opinions toward complex claims, proposals or policy statements.

Conclusion

1. Limitations

- **Conversational Complexity:** Jev struggles to interpret stance in multi-turn conversations.
- **Stance Direction:** Higher confusion between Favor and Against in ZS-CSD.
- **Target Dependency:** Claim-based targets are more challenging than noun-phrase targets.
- **Cross-Dataset Performance:** Performance varies between individual-text and conversational datasets.

2. Key Takeaways

- Jev achieves competitive Macro-F1 on VAST (77.92%).
- Jev outperforms the tested Flash models on both datasets.
- GPT-5.6 performs better on ZS-CSD (73.62%).
- Jev demonstrates substantially lower API cost in the reported experiments.

Thank you